

Framing Women with the Pronoun “她” in Republican-Era Newspapers (1911–1949)

MACHINE-LEARNING EVIDENCE FROM HEADLINE SENTIMENT & TOPICS

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Date: 2025.10.05

INTRODUCTION

01

During the Republican era (1911 – 1949), the pronoun “她” entered print, making women newly visible. This poster tests whether the 1937 full-scale war shifted the tone of headlines that use “她,” via sentiment analysis and DiD/event-study on 1911 – 1949 titles; this is crucial to understanding whether visibility meant better portrayals or merely a lexical shift, with relevance to today’s media.

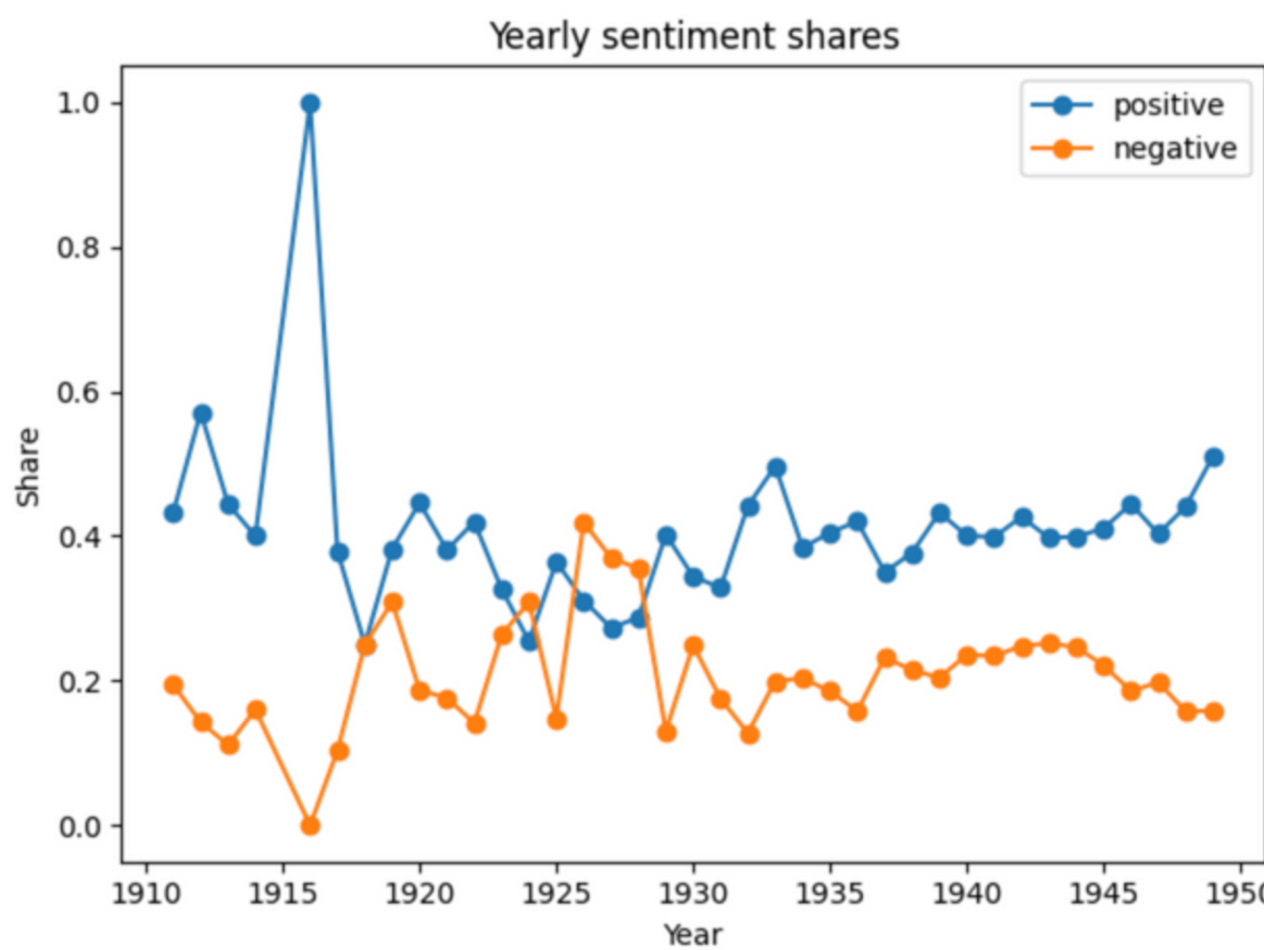
Keywords: Republican China; gender; wartime media; sentiment analysis; difference-in-differences; mobilization; language and visibility

METHODS

02

- Data:** Shanghai Library National Periodical Index headlines (titles only), 1911 – 1949. Keep year and paper; flag has_ta if the title explicitly contains “她”. Split pre-1937 vs post-1937.
- Sentiment analysis:** We used SentProp (seeded-lexicon) to score tokens, aggregated scores to a headline-level polarity, and classified each title as Positive /Neutral /Negative
- Regression:** To further characterize the effect, I estimate: (1) an event study tracing the yearly positivity gap around 1937 to assess pre-trends and dynamics; (2) a segmented DiD summarizing the average positivity gap in 1937 – 41 and 1942 – 45 versus pre-1937; and (3) a segmented dose-response DDD testing whether years with higher mobilization display a larger positivity gap.

$$\Pr(\text{Positive}_{it}=1)=\alpha+\beta(\text{has_ta}_i\times\text{Post1937}_t)+\mu_{\text{paper}}+\lambda_y+\varepsilon_{it}\text{ (Main Regression)}$$



DISCUSSION

03

- After 1937, overall sentiment in headlines becomes more positive, with the share of positive articles rising from 39% to 43% and neutral ones declining from 40% to 37%.
- Regression results confirm this pattern: female-related headlines are about +0.19 points more positive in the early war (1937 – 41), while no significant difference appears after 1942.
- When mobilization intensity is considered, the average effect across all war years is small, but the interaction between female coverage and mobilization during 1942 – 45 is +0.22 per 1 SD increase in mobilization.
- Taken together, these findings suggest that media tone turned more favorable after the onset of total war, with women portrayed especially positively at the beginning of the conflict and again in high-mobilization years—indicating that mobilization, rather than war itself, sustained positive gender portrayals.

